

STRING MATCHING ALGORITHM

MODULE-3

Rabin-Karp Algorithm

- Rabin-Karp is another pattern searching algorithm to find the pattern in a more efficient way.
- It also checks the pattern by moving window one by one, but without checking all characters for all cases, it finds the hash value.
- When the hash value is matched, then only it tries to check each character.
- This procedure makes the algorithm more efficient.
- The Rabin-Karp string matching algorithm calculates a hash value for the pattern, as well as for each M-character subsequences of text to be compared.
- If the hash values are unequal, the algorithm will determine the hash value for next M-character sequence.
- If the hash values are equal, the algorithm will analyze the pattern and the M-character sequence.
- In this way, there is only one comparison per text subsequence, and character matching is only required when the hash values match.

Rabin-Karp Algorithm

RABIN-KARP-MATCHER (T, P, d, q)

1. $n \leftarrow \text{length}[T]$
2. $m \leftarrow \text{length}[P]$
3. $h \leftarrow d^{m-1} \bmod q$
4. $p \leftarrow 0$
5. $t0 \leftarrow 0$
6. **for** $i \leftarrow 1$ **to** m //Preprocessing
7. **do** $p \leftarrow (dp + P[i]) \bmod q$
8. $t0 \leftarrow (dt0 + T[i]) \bmod q$
9. **for** $s \leftarrow 0$ **to** $n-m$ //Matching
10. **do if** $p = ts$
11. **then if** $P[1.....m] = T[s+1.....s + m]$
12. **then** "Pattern occurs with shift" s
13. **if** $s < n-m$
14. **then** $ts+1 \leftarrow (d(ts - T[s+1]h) + T[s+m+1]) \bmod q$

Example: pattern length is 3 and string is "23456"

You compute the value of the first window (which is "234") as 234.

How will you compute the value of the next window "345"? You will do

$(234 - 2*100)*10 + 5$ and get 345.

$$=(2*10^2+3*10^1+4*10^0)$$

$$=(200+30+4)=234(\text{hash code})$$

Hash code for next shift

$$=((234-200)*10)+5=345$$

Problem

Explain spurious hits in Rabin-Karp string matching algorithm with an example. Working modulo $q = 13$, how many spurious hits does the Rabin-Karp matcher encounter in the text $T = 2359023141526739921$ when looking for the pattern $P = 31415$?

$T = 2359023141526739921$

$P = 31415$

$n=19, m=5, q=13, d=10$

$h \leftarrow d^{m-1} \bmod q$

$h = 10^5 - 1 \bmod 13$

$h=3$

$P=0$

$t_0=0$

for i= 1 to 5

$p \leftarrow (dp + P[i]) \bmod q$

i=1

$p = (10^0 + P[1]) \bmod 13 \Rightarrow p = 3$

i=2

$p = (10^3 + P[2]) \bmod 13 \Rightarrow p = 5$

i=3

$p = (10^5 + P[3]) \bmod 13 \Rightarrow p = 2$

i=4

$p = (10^2 + P[4]) \bmod 13 \Rightarrow p = 8$

i=5

$p = (10^8 + P[5]) \bmod 13 \Rightarrow p = 7$

$t0 \leftarrow (dt0 + T[i]) \bmod q$

$t0 = (10^0 + T[1]) \bmod 13 \Rightarrow t0 = 2$

2 mod 13

$t0 = (10^2 + T[2]) \bmod 13 \Rightarrow t0 = 10$

23 mod 13

$t0 = (10^{10} + T[3]) \bmod 13 \Rightarrow t0 = 1$

105 mod 13

$t0 = (10^1 + T[4]) \bmod 13 \Rightarrow t0 = 6$

19 mod 13

$t0 = (10^6 + T[5]) \bmod 13 \Rightarrow t0 = 8$

60 mod 13

$S < -0$ to 14 , $p=7, t_0=8, h=3, q=13, m=5$

if $p = t_s$

if $s < n-m$, $t_{s+1} \leftarrow (d(t_s - T[s+1]h) + T[s+m+1]) \bmod q$

$S=0$ $7=8$

$0 < 14$, $t_{0+1} = (10 * (8 - T[1] * 3) + T[0+5+1]) \bmod 13$

$t_1 = (10 * (8 - 2 * 3) + 2) \bmod 13$

$t_1 = (10 * (8 - 6) + 2) \bmod 13$

$t_1 = (10 * (2) + 2) \bmod 13$

$t_1=9$

$S=1$ $7=9$

$1 < 14$, $t_2 = (10 * (9 - T[2] * 3) + T[1+5+1]) \bmod 13 \Rightarrow$ **$t_2=13$**

$S=2$ $7=13$

$2 < 14$, $t_3 = (10 * (13 - T[3] * 3) + T[2+5+1]) \bmod 13 \Rightarrow$ **$t_3=11$**

$S=3$ $7=11$

$3 < 14$, $t_4 = (10 * (11 - T[4] * 3) + T[3+5+1]) \bmod 13 \Rightarrow$ **$t_4=0$**

$S=4$ $7=0$

$4 < 14$, $t_5 = (10 * (0 - T[5] * 3) + T[4+5+1]) \bmod 13 \Rightarrow$ **$t_5=1$**

$S=5$ $7=1$

$5 < 14$, $t_6 = (10 * (1 - T[6] * 3) + T[5+5+1]) \bmod 13 \Rightarrow$ **$t_6=7$**

$S < -0 \text{ to } 14, p=7, t6=7, h=3, q=13, m=5$

if $p = ts$

if $s < n-m, ts+1 \leftarrow (d(ts-T[s+1]h)+T[s+m+1]) \bmod q$

$S=6 \quad 7=7$

$6 < 14, t2=(10*(7-T[7]*3)+T[6+5+1]) \bmod 13 \Rightarrow t7=8$

$P[1.....m] = T[s+1.....s+m]$

$P[1]=T[7] \Rightarrow 3=3$

$P[2]=T[8] \Rightarrow 1=1$

$P[3]=T[9] \Rightarrow 4=4$

$P[4]=T[10] \Rightarrow 1=1$

$P[5]=T[11] \Rightarrow 5=5$

TRUE

then "Pattern occurs with shift" s

"Pattern occurs with shift" **6**

$S < -0$ to 14 , $p=7, t_7=8, h=3, q=13, m=5$

if $p = ts$

if $s < n-m$, $ts+1 \leftarrow (d(ts-T[s+1]h)+T[s+m+1]) \bmod q$

$S=7$ $7=8$

$7 < 14$, $t_8 = (10 * (8 - T[8] * 3) + T[7+5+1]) \bmod 13 \Rightarrow t_8=4$

$S=8$ $7=4$

$8 < 14$, $t_9 = (10 * (4 - T[9] * 3) + T[8+5+1]) \bmod 13 \Rightarrow t_9=5$

$S=9$ $7=5$

$9 < 14$, $t_{10} = (10 * (5 - T[10] * 3) + T[9+5+1]) \bmod 13 \Rightarrow t_{10}=10$

$S=10$ $7=10$

$10 < 14$, $t_{11} = (10 * (10 - T[11] * 3) + T[10+5+1]) \bmod 13 \Rightarrow t_{11}=11$

$S=11$ $7=11$

$11 < 14$, $t_{12} = (10 * (11 - T[12] * 3) + T[11+5+1]) \bmod 13 \Rightarrow t_{12}=7$

$S=12$ $7=7$

$12 < 14$, $t_{13} = (10 * (7 - T[13] * 3) + T[12+5+1]) \bmod 13 \Rightarrow t_{13}=9$

$P[1]=T[13] \Rightarrow 2=6$ (FALSE) $\rightarrow$ spurious hits

$S=13$ $7=9$

$13 < 14$, $t_{14} = (10 * (9 - T[14] * 3) + T[13+5+1]) \bmod 13 \Rightarrow t_{14}=11$

$S=14$ $7=11$

$14 < 14$

Time Complexity:

- The running time of **RABIN-KARP-MATCHER** in the worst case scenario is **$O((n-m+1)m)$** i.e. **$O(mn)$** , The worst-case complexity occurs when spurious hits occur a number for all the windows.
- If the expected number of strong shifts is small and prime q is chosen to be quite large, then the Rabin-Karp algorithm can be expected to run in time **$O(n+m)$** (Best and Average case)

Practice

1.) working module $q = 11$, how many spurious hits does the Rabin-Karp matcher encounters in Text $T = 31415926535$, $P=26$

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OUTPUT:

Pattern occurs with shift 6